TRI-OCTAGON MATHEMATICAL RESEARCH

Capacity-Sized Receiving and Native Recovery in the Reciprocal Chiral Exchange Candidate

A same-trajectory finite-parameter certificate
for coherent empty launches

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Abstract

Finite-time access to the joint projection capacity of the reciprocal chiral exchange candidate does not by itself determine receiver allocation or later native recovery. We establish both on the same unchanged solution for the coherent empty launch $\Omega_0 = (1, e^{i\phi}, 1)$, $0 < \phi \le 1/100$ radians, at yaw $\pi/4$. Fix finite positive coefficients before phase so that $\omega_R = \nu/3$ and $0 < \gamma/\nu \le 1/100$, with the rates defined in the original variables below. The admitted periodic carrier retains positive propagation and disjoint finite-width H^1 profiles. Writing $C = \alpha_N(1 - \cos\phi)$ for the joint capacity and $\varpi = \sqrt{17/18}\,\nu$, the episode satisfies

$$E_{\mathrm{out}}(1/\nu) > \frac{1053}{4096}C, \qquad H_R(\pi/\varpi) > C/4,$$
$$H_C(\pi/\varpi) < C/100, \qquad E_{\mathrm{out}}(2\pi/\varpi) < C/100.$$

The first bound is inherited after checking the same coefficients. Net native recovery to $2\pi/\varpi$ exceeds $24/25$ of each actual intermediate external stock, at $1/\nu$ and π/ϖ, respectively. A complete moving-native tangent reference and a uniform finite error below $\sqrt{2C}/10$ prove the new stock bounds. The normalization is joint capacity, not the whole initial native store; absolute capacity tends to zero with phase. The equality-constrained sufficient family proves neither saturation, irreversible retention, monotonic recovery, an exact nonlinear period nor a spatial circuit.

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1 From accessible capacity to receiving and recovery

The reciprocal chiral exchange candidate couples an evolving native state to a periodic carrier and a conservative collocated receiver [1]. Its joint projection ceiling bounds the trial storage that can leave the native state [3]. The finite-capacity access theorem then supplies an actual external stock larger than a quarter of that ceiling at an explicit finite time [4]. That access result leaves two separate questions: how much of the stock reaches the receiver, and whether the native store subsequently recovers a quantified amount on the same solution.

Earlier receiving and recovery guarantees use a different sufficient coefficient regime [2, 3]. Section 4 shows why those certificates cannot simply be appended to the access theorem. We instead use a tangent comparison in which the native mode, carrier and receiver all move. Its full finite-error estimate controls the exact nonlinear solution with positive propagation throughout the episode. This paper presents the accepted receiving/recovery investigation [5, 6]; it introduces no new model, parameter optimization or physical interpretation.

The result is deliberately restricted. Coefficients are fixed before every phase in a small interval, and the relation between native and receiver rates is an equality. The theorem is a certificate for this nonempty subfamily, not an open robustness neighborhood or a classification of all couplings. *Capacity-sized* always refers to the joint ceiling, which itself becomes small near the aligned launch.

For clarity, the four predecessor results with the same theorem number have distinct roles. P1 Theorem 3.1 supplies global existence and strong empty-launch regularity; P2 Theorem 3.1 is the older native-credit result; P3 Theorem 3.1 supplies projection capacity; and P4 Theorem 3.1 supplies finite-time access. Hereafter P1–P4 mean references [1, 2, 3, 4], respectively.

2 Fixed model and inherited accounts

2.1 Raw registration, finite ports and original stores

Write $\alpha = \alpha_N > 0$, $\beta = \beta_R > 0$, $\eta = \eta_N > 0$ and $\kappa = \eta_R > 0$ for the original coefficients. All are finite and constant in exchange time. Keep the original positive registration gains, $G = \kappa_{\mathrm{in}}\lambda_g$, and fix yaw $\delta_g = \pi/4$. With raw two-port order $+, -$,

$$\begin{gathered} x = (a, b), \quad \Omega = a + ib, \quad f(x) = K(a \times b), \quad J(x) = D_x f(x), \\ K = \frac{G}{\sqrt{7}} \begin{pmatrix} \sqrt{2} & \sqrt{2} & \sqrt{3} \\ \sqrt{2} & \sqrt{2} & -\sqrt{3} \end{pmatrix}, \qquad k = \|K\| = G\sqrt{8/7}. \end{gathered} \tag{2.1}$$

The raw source signs are not changed by ambient contact orientations.

Retain the admitted circle $s \in \mathbb{R}/L_c\mathbb{Z}$ with $0 < L_c < \infty$ and $c > 0$. Contacts $s_\pm = L_c/2 \mp \eta_{\mathrm{geom}}$ have whole-circle shortest separation $2\eta_{\mathrm{geom}} \le L_c/2$. For $0 < \epsilon_c < \eta_{\mathrm{geom}}$,

$$\begin{gathered} b_\sigma(s) = \epsilon_c^{-1}\left(1 - \frac{|\operatorname{wrap}_{L_c}(s - s_\sigma)|}{\epsilon_c}\right)_+, \quad \mathcal{B}q = b_+ q_+ + b_- q_-, \\ \mathcal{B}^* p = \left(\int b_+ p \,\mathrm{d}s, \int b_- p \,\mathrm{d}s\right)^T, \qquad m = \frac{2}{3\epsilon_c}, \quad d = \frac{2}{\epsilon_c^3}, \\ \mathcal{B}^*\mathcal{B} = mI_2, \qquad \|\mathcal{B}\| = \|\mathcal{B}^*\| = \sqrt{m}, \qquad \|\partial_s \mathcal{B} q\|_2 = \sqrt{d}\,|q|. \end{gathered} \tag{2.2}$$

All integrals cover one whole circle. The separation assumption ensures disjoint supports across the periodic seam as well as locally. Integrating the two affine halves of a continuous tent gives (2.2); disjointness removes derivative cross terms. These profiles belong to periodic H^1. Their first weak derivatives lie in L^2, with no delta terms; no second L^2 derivative is assumed.

The unchanged equations and trial stores are [1, Section 1]

$$\boxed{\begin{aligned} \dot{x} &= -\frac{\eta}{2\alpha} J(x)^T y, & y &= \mathcal{B}^* p, \\ p_t + c(r_c)_s &= \mathcal{B}\{\eta f(x) - \kappa z\}, & (r_c)_t + c p_s &= 0, \\ \dot{z} &= \frac{\kappa}{\beta} y. \end{aligned}} \tag{2.3}$$

$$\begin{aligned} H_N &= \frac{\alpha}{2}(|a|^2 + |b|^2), & H_C &= \frac{1}{2}(\|p\|_2^2 + \|r_c\|_2^2), \\ H_R &= \frac{\beta}{2}|z|^2, & E_{\text{out}} &= H_C + H_R. \end{aligned} \tag{2.4}$$

All conclusions below use these original stores. The auxiliary norm introduced later is a different proof device.

2.2 Strong evolution and the exact account

P1 Theorem 3.1 supplies global existence and uniqueness before any new estimate. Empty carrier preparation belongs to H^1_{per}, hence

$$(p, r_c) \in C(\mathbb{R}; H^1_{\text{per}})^2 \cap C^1(\mathbb{R}; L^2)^2, \qquad a, b, z \in C^1. \tag{2.5}$$

The adjoint port is bounded, so $y = \mathcal{B}^* p$ is continuous. For $w = K^T y$ and $\mu = \eta/(2\alpha)$, differentiation gives

$$J^T y = (b \times w, w \times a), \qquad \dot{a} = \mu w \times b, \qquad \dot{b} = -\mu w \times a. \tag{2.6}$$

The homogeneous identity $J(x)x = 2f(x)$ and periodic integration of the carrier product derivative give the original accounts [1, Proposition 2.1]:

$$\begin{aligned} P_{NC} &= \eta f(x)^T y, & P_{CR} &= \kappa z^T y, \\ \dot{H}_N &= -P_{NC}, & \dot{H}_C &= P_{NC} - P_{CR}, & \dot{H}_R &= P_{CR}, \\ H_N(t) + H_C(t) + H_R(t) &= H_N(0), & E_{\text{out}}(t) &= H_N(0) - H_N(t). \end{aligned} \tag{2.7}$$

The last line uses empty external preparation, as imposed below. All powers are continuous. This argument uses the periodic H^1 product rule, not pointwise second derivatives.

3 A same-trajectory episode certificate

Work first in the published normalized native-amplitude convention, so $R = \sqrt{3}$. Define

$$\begin{aligned} \nu^2 &= \frac{\eta^2 m k^2 R^2}{2\alpha}, & \sigma_{\text{ext}}^2 &= \frac{c^2 d}{m} + \frac{\kappa^2 m}{\beta}, \\ \gamma &= c\sqrt{d/m}, & \omega_R &= \kappa\sqrt{m/\beta}, & \omega_N &= \sqrt{5/6}\,\nu, \quad \varpi = \sqrt{17/18}\,\nu. \end{aligned} \tag{3.1}$$

First fix the admitted geometry, positive storage scales and one finite positive coefficient tuple satisfying

$$\omega_R = \nu/3, \qquad 0 < \gamma/\nu \le 1/100. \tag{3.2}$$

For each phase, prepare the exact coherent empty launch

$$0 < \phi \le 1/100, \qquad \Omega_0 = (1, e^{i\phi}, 1), \qquad p_0 = r_{c,0} = z_0 = 0. \tag{3.3}$$

The angle is in radians. This interval lies inside P4's $(0, \pi/12)$; an elementary bound on π is given in Section 7.1. P3 Theorem 3.1 and its strict yaw witness give

$$\begin{aligned} C = C_{\text{joint}} = \alpha(1-\cos\phi) > 0, \qquad & 0 \le E_{\text{out}}(t) \le C \quad (t \ge 0), \\ f_{0,\pm} = \frac{G\sin\phi}{\sqrt{7}}(-\sqrt{2} \pm \sqrt{3}), \qquad & |f_0|^2 = \frac{10G^2\sin^2\phi}{7} > 0. \end{aligned} \tag{3.4}$$

Nonnegative stores and conservation also give $|x(t)| \le R$. Neither this native ball nor the ceiling assumes that the coherent launch family remains invariant during evolution.

Theorem 3.1 (Capacity-sized receiving and net recovery)**.** *For every fixed tuple satisfying* (3.2) *and every phase in* (3.3)*, the phase's unique actual solution of* (2.3) *has*

$$0 < t_A = 1/\nu < t_R = \pi/\varpi < t_F = 2\pi/\varpi. \tag{3.5}$$

At these times,

$$\begin{gathered} E_A := E_{\text{out}}(t_A) > \frac{1053}{4096}C, \\ \boxed{H_R(t_R) > C/4, \qquad H_C(t_R) < C/100, \qquad E_F := E_{\text{out}}(t_F) < C/100.} \end{gathered} \tag{3.6}$$

The first bound is P4's access theorem applied after checking the same tuple in Section 8. The boxed bounds are the new finite-error conclusions.

With $E_R = E_{\text{out}}(t_R)$, all denominators below are positive, and

$$\begin{aligned} \frac{25}{26} < \frac{H_R(t_R)}{E_R} \le 1, \qquad & 0 \le \frac{H_R(t_F)}{H_R(t_R)} < \frac{1}{25}, \\ \frac{24}{25} < \mathcal{R}_A := \frac{H_N(t_F) - H_N(t_A)}{E_A} \le 1, \qquad & \\ \frac{24}{25} < \mathcal{R}_R := \frac{H_N(t_F) - H_N(t_R)}{E_R} \le 1, \qquad & H_N(t_F) - H_N(t_R) > \frac{6}{25}C. \end{aligned} \tag{3.7}$$

Sections 5–8 prove the result. The coefficients and all three times are phase independent at one fixed tuple; different phases are separately prepared solutions. No parameter is switched along any trajectory. Section 9 proves that the finite positive family is nonempty.

For notation, E_R is combined external stock at t_R, whereas $H_R(t_R)$ is receiver stock alone. The reference coordinates X, P, Z used later are weighted amplitudes, distinct from the powers P_{NC}, P_{CR} and any native bilinear invariant. The ratios in (3.7) concern stocks and net changes between stated times, not a tagged energy cohort.

4 Why the older certificates cannot simply be joined

The older transfer theorem, P3 Theorem 4.1, uses a sufficient receiver threshold depending on a tolerance $0 < q \le 1/5$. In this section only, use P3's notation

$$\begin{gathered} \omega = \sqrt{m/\beta}, \quad S = 2\pi/\omega, \quad \Lambda = \sqrt{\beta}\,|f_0|, \quad F_{\max} = kR^2/2, \\ A = \nu^2 F_{\max}/2, \qquad B_2 = \sqrt{m}\,AS^3/3. \end{gathered} \tag{4.1}$$

The inherited bound $|f(x)| \le k|x|^2/2$ gives $F_{\max} \ge |f_0| > 0$. The B_2 part of P3's sufficient threshold, its equation (4.3), requires

$$\begin{aligned} \kappa^2 \ge \frac{2B_2}{q\Lambda} \implies \left(\frac{\kappa\omega}{\nu}\right)^2 &\ge \frac{8\pi^3 F_{\max}}{3q|f_0|} \\ &\ge \frac{8\pi^3}{3q} > \frac{1}{4}. \end{aligned} \tag{4.2}$$

Every divided quantity is positive, and the substitution uses $\sqrt{m}/\sqrt{\beta} = \omega$. The elementary bound $\pi > 1$ and $1/q \geq 5$ already make the final lower bound exceed $40/3$.

P4's sufficient condition instead requires $\gamma^2 + (\kappa\omega)^2 \leq \nu^2/4$. Thus these sufficient certificate regimes are disjoint. This does not prove impossibility of simultaneous access, receiving and recovery behaviors. We do not import P3's $80/81$ fraction or the exact-time returning-power witness of P2 Theorem 3.1 into the present tuple. A new finite-error bridge is required.

5 Tangent geometry and the exact nonlinear remainder

5.1 Rank-one tangent of the unchanged registration

Let $v = (1,1,1)$, $e = (1,-1,0)/\sqrt{2}$, $u = a - v$, and $[v]_\times q = v \times q$. Direct multiplication gives

$$\begin{aligned} D = K[v]_\times &= \frac{G}{\sqrt{7}} \begin{pmatrix} \sqrt{2}-\sqrt{3} & -\sqrt{2}+\sqrt{3} & 0 \\ \sqrt{2}+\sqrt{3} & -\sqrt{2}-\sqrt{3} & 0 \end{pmatrix} = \ell e^T, \\ \ell = De &= \frac{G}{\sqrt{7}} \begin{pmatrix} 2-\sqrt{6} \\ 2+\sqrt{6} \end{pmatrix}, \qquad |\ell|^2 = \frac{20G^2}{7}, \qquad Ke = 0, \\ \omega_N^2 &= \frac{\eta^2 m |\ell|^2}{2\alpha} = \frac{5}{6}\nu^2. \end{aligned} \tag{5.1}$$

The tangent drive D has rank one; the exact K has rank two. In particular, the full drive remains $f = Db + K(u \times b)$.

5.2 A weighted forced estimate

For $Y = (u, b, p, r_c, z)$, introduce the Hilbert norm

$$\|Y\|_*^2 = 2\alpha(|u|^2 + |b|^2) + \|p\|_2^2 + \|r_c\|_2^2 + \beta|z|^2. \tag{5.2}$$

This is an auxiliary proof norm, not the original native store or a newly conserved nonlinear energy. The finite shift preserves the strong regularity already supplied by P1.

On the domain with periodic H^1 carrier coordinates and unrestricted finite coordinates, define

$$\mathcal{A}_c Y = \left(0, -\mu D^T y,\ -c(r_c)_s + \eta \mathcal{B} D b - \kappa \mathcal{B} z,\ -c p_s,\ \frac{\kappa}{\beta} y\right). \tag{5.3}$$

Since $D^T y = w \times v$, the weighted tangent native term is $-2\alpha\mu(Db)^T y = -\eta(Db)^T y$. The complete cancellation is

$$\begin{aligned} \langle Y, \mathcal{A}_c Y \rangle_* = &-\eta(Db)^T y + \eta y^T D b \\ &- c\int [p(r_c)_s + r_c p_s]\,\mathrm{d}s - \kappa y^T z + \kappa z^T y = 0. \end{aligned} \tag{5.4}$$

The periodic H^1 product rule removes the spatial integral. The native weight 2α is essential: using α would leave an uncancelled tangent term.

For an already strong curve $\mathcal{Z}$ satisfying $\dot{\mathcal{Z}} = \mathcal{A}_c \mathcal{Z} + F$, differentiate $(\|\mathcal{Z}\|_*^2 + \delta^2)^{1/2}$ with $\delta > 0$. Equation (5.4) and Cauchy–Schwarz bound its derivative above by $\|F\|_*$. Integrating and letting $\delta \downarrow 0$ gives

$$\|\mathcal{Z}(t)\|_* \leq \|\mathcal{Z}(0)\|_* + \int_0^t \|F(s)\|_*\,\mathrm{d}s, \qquad t \geq 0. \tag{5.5}$$

This includes zero norm without dividing by it. Existence is inherited before this calculation, not inferred from skew cancellation.

5.3 Complete nonlinear decomposition

Using (2.6) and $f = Db + K(u \times b)$ gives the exact equation

$$\begin{aligned} \dot{Y} &= \mathcal{A}_c Y + \mathcal{N}(Y), \\ \mathcal{N}(Y) &= (\mu w \times b, -\mu w \times u, \eta \mathcal{B} K(u \times b), 0, 0)\,. \end{aligned} \tag{5.6}$$

No quadratic term is discarded. Put $X_N = \sqrt{2\alpha}\sqrt{|u|^2 + |b|^2}$ and $V_{\mathrm{ext}}^2 = \|p\|_2^2 + \|r_c\|_2^2 + \beta|z|^2$. Since $|w| \le k\sqrt{m}\, V_{\mathrm{ext}}$, the weighted native remainder is at most $LX_N V_{\mathrm{ext}}$, while the external remainder is at most $\eta k\sqrt{m}\,|u||b| \le (L/2)X_N^2$, where

$$\begin{aligned} L &= \frac{\eta k \sqrt{m}}{2\alpha}, \\ \|\mathcal{N}(Y)\|_*^2 &\le L^2(X_N^2 V_{\mathrm{ext}}^2 + X_N^4/4) \le L^2(X_N^2 + V_{\mathrm{ext}}^2)^2, \\ \|\mathcal{N}(Y)\|_* &\le L\,\|Y\|_*^2\,. \end{aligned} \tag{5.7}$$

The blocks are orthogonal in (5.2). The second squared inequality has gap $L^2(3X_N^4/4 + X_N^2 V_{\mathrm{ext}}^2 + V_{\mathrm{ext}}^4) \ge 0$, proving the stated global bound.

6 Exact initial data and a moving-native reference

6.1 Positive scalar majorant

The exact initial deviations, with no first-order phase approximation, are

$$\begin{aligned} u_0 &= (0, \cos\phi - 1, 0), \qquad b_0 = (0, \sin\phi, 0), \\ \|Y(0)\|_*^2 &= 2\alpha[(\cos\phi - 1)^2 + \sin^2\phi] = 4C. \end{aligned} \tag{6.1}$$

Write $y_0 = 2\sqrt{C} > 0$. The known global strong solution and (5.5)–(5.7) give $\|Y(t)\|_* \le y_0 + L\int_0^t \|Y(s)\|_*^2\,\mathrm{d}s =: Q(t)$. On every finite interval, Q is positive and continuously differentiable, with $Q(0) = y_0$ and $Q' \le LQ^2$. Therefore $(1/Q)' = -Q'/Q^2 \ge -L$, and $1/Q(t) \ge 1/y_0 - Lt$. Only while this lower bound is positive may it be inverted:

$$\|Y(t)\|_* \le Q(t) \le \frac{y_0}{1 - Ly_0 t}, \qquad 0 \le t < (Ly_0)^{-1}. \tag{6.2}$$

Section 7.1 proves directly that $Ly_0 t_F < 1/25$ before this estimate is used up to t_F. There is no assumed bootstrap or new existence construction.

6.2 Full reference, including static components

Use $\dot{Y}^{(0)} = \mathcal{A}_0 Y^{(0)}$ with precisely the initial data (6.1). Only the reference removes propagation and the quadratic remainder; actual c and κ stay positive.

Set $n = \ell/|\ell|$, $h_0 = e^T b_0 = -\sin\phi/\sqrt{2}$, and $b_\perp = b_0 - h_0 e = (\sin\phi/2, \sin\phi/2, 0)$. Then $Db_\perp = 0$. The complete reference is

$$\begin{aligned} u^{(0)}(t) &= u_0, \qquad b^{(0)}(t) = b_\perp + eX(t)/\sqrt{2\alpha}, \\ p^{(0)}(t) &= \mathcal{B}n\,P(t)/\sqrt{m}, \quad r_c^{(0)}(t) = 0, \quad z^{(0)}(t) = nZ(t)/\sqrt{\beta}. \end{aligned} \tag{6.3}$$

The real deviation, including its component parallel to e, and the inactive imaginary component are retained. Because $D^T n = |\ell| e$ and $y^{(0)} = \sqrt{m}\,nP$, all blocks of $\mathcal{A}_0$ reduce to

$$\begin{aligned} \dot{X} &= -\omega_N P, \qquad \dot{P} = \omega_N X - \omega_R Z, \qquad \dot{Z} = \omega_R P, \\ X_0 &= -\sqrt{\alpha}\sin\phi < 0, \qquad P_0 = Z_0 = 0. \end{aligned} \tag{6.4}$$

The negative sign follows from the exact phase launch, with $0 < \phi < \pi/2$. The solution is

$$\begin{aligned} X(t) &= X_0 \frac{\omega_R^2 + \omega_N^2 \cos(\varpi t)}{\varpi^2}, \\ P(t) &= X_0 \frac{\omega_N}{\varpi} \sin(\varpi t), \\ Z(t) &= X_0 \frac{\omega_N \omega_R}{\varpi^2} [1 - \cos(\varpi t)]. \end{aligned} \tag{6.5}$$

Differentiation verifies the three ODEs and all initial data, and $X^2 + P^2 + Z^2 = X_0^2$. Consequently,

$$\begin{aligned} (p^{(0)}, r_c^{(0)})(t_R) = 0, \qquad & (p^{(0)}, r_c^{(0)}, z^{(0)})(t_F) = 0, \\ \frac{X_0^2}{2} = C \cos^2(\phi/2), \qquad & \frac{|Z(t_R)|}{\sqrt{2C}} = \frac{2\sqrt{30}}{17} \cos(\phi/2). \end{aligned} \tag{6.6}$$

In particular, $|X_0|/\sqrt{2C} = \cos(\phi/2) \le 1$, with positive cosine. This reference is a strong curve: a fixed H^1 profile multiplies a smooth scalar. Its period is not asserted to be an actual nonlinear period, and the actual trajectory is not restarted at either landmark.

7 Uniform finite error on the complete episode

7.1 Positive denominator and exact constants

The elementary numerical bounds used here have exact integral certificates:

$$\begin{aligned} \frac{22}{7} - \pi &= \int_0^1 \frac{s^4(1-s)^4}{1+s^2} \, \mathrm{d}s > 0, \\ \frac{1}{1+s^2} - (1 - s^2) &= \frac{s^4}{1+s^2} > 0 \quad (0 < s < 1), \qquad \pi > 8/3. \end{aligned} \tag{7.1}$$

For the first identity, polynomial division gives quotient $s^6 - 4s^5 + 5s^4 - 4s^2 + 4$, with integral $22/7$, and remainder -4. Integrating $4/(1+s^2)$ gives π. For the lower bound integrate the second identity and use $\int_0^1 (1 - s^2)\, \mathrm{d}s = 2/3$. Strict positivity on $(0, 1)$ proves both inequalities. Thus $1/100 < \pi/12$ and $\pi > 1$.

Put $T = t_F$ and $\theta = Ly_0 T = 2L\sqrt{C}\, T$. Since $1 - \cos\phi = 2\sin^2(\phi/2) \le \phi^2/2$,

$$\begin{aligned} \nu T &= 2\pi \sqrt{18/17} < 20/3, \qquad 1/\sqrt{3} < 3/5, \\ \theta &= \frac{\sqrt{2}}{R} \sqrt{1 - \cos\phi}\, \nu T \le \frac{\phi}{\sqrt{3}} \nu T < 4\phi \le \frac{1}{25}. \end{aligned} \tag{7.2}$$

The time comparison follows from $\pi < 22/7$ and $(20/3)^2 - (44/7)^2(18/17) = 19568/7497 > 0$. Similarly $(3/5)^2 - 1/3 = 2/75 > 0$. The sine bound follows by integrating cosine. Therefore $1 - Ly_0 t \ge 1 - \theta > 24/25$ throughout $[0, T]$, establishing the positive denominator before using (6.2) there.

Chronology also follows: $\varpi/\nu = \sqrt{17/18} < 1 < \pi$, so $t_A < t_R$, while $t_F = 2t_R > t_R > 0$. These are comparison landmarks for evaluating the actual solution, not inferred physical cycle times.

7.2 Strong difference before estimating its residual

The actual curve is strong by P1 and the finite shift. The reference is strong by (6.3)–(6.5). Their difference $E = Y - Y^{(0)}$ therefore has $C_t H^1$ carrier coordinates and is C_t^1 in the Hilbert space before a residual is estimated. Subtracting the two equations gives the exact identity

$$\dot{E} = \mathcal{A}_c E + \mathcal{N}(Y) + (0, 0, 0, -c\partial_s p^{(0)}, 0), \qquad E(0) = 0. \tag{7.3}$$

The first carrier residual vanishes because $r_c^{(0)} = 0$; the second is $-c\partial_s p^{(0)}$. Native and receiver interactions remain in both operators. No receiver term is omitted. Only the first weak spatial derivative is used; no second tent derivative or operator-norm convergence of an unbounded generator is invoked.

Apply (5.5) to (7.3). Equations (5.7), (6.2) and (7.2) give

$$\begin{aligned}\int_0^T \|\mathcal{N}(Y(t))\|_* \,\mathrm{d}t &\le \int_0^T \frac{Ly_0^2}{(1-Ly_0t)^2}\,\mathrm{d}t = \frac{y_0\theta}{1-\theta},\\ \int_0^T c\left\|\partial_s p^{(0)}(t)\right\|_2 \,\mathrm{d}t &= \int_0^T \gamma|P(t)|\,\mathrm{d}t = \frac{4\gamma|X_0|\omega_N}{\varpi^2}\\ &\le \sqrt{2C}\frac{4\gamma\omega_N}{\varpi^2}.\end{aligned} \tag{7.4}$$

Here $\|\partial_s \mathcal{B}n\|_2 = \sqrt{d}$, and $\int_0^{2\pi} |\sin q|\,\mathrm{d}q = 4$ by splitting the two signed half-periods. The integrands in both estimates are nonnegative; their full-T integrals bound every earlier time. Division by the positive $\sqrt{2C}$ yields

$$\sup_{0\le t\le T} \frac{\|E(t)\|_*}{\sqrt{2C}} \le \frac{\sqrt{2}\,\theta}{1-\theta} + \frac{4\gamma\omega_N}{\varpi^2}. \tag{7.5}$$

7.3 Strict margins on the allowed endpoints

Using the exact rate ratios,

$$\frac{4\gamma\omega_N}{\varpi^2} = \frac{72}{17}\sqrt{5/6}\frac{\gamma}{\nu} < 4\frac{\gamma}{\nu} \le \frac{1}{25},$$

because $16 - (72/17)^2(5/6) = 304/289 > 0$. Also $\sqrt{2} < 10/7$, since $(10/7)^2 - 2 = 2/49 > 0$. The function $\theta/(1-\theta)$ is increasing on $[0,1)$; hence the strict $\theta < 1/25$ gives

$$\begin{aligned}\sup_{0\le t\le t_F} \frac{\|E(t)\|_*}{\sqrt{2C}} &< \frac{10}{7}\frac{1}{24} + \frac{1}{25} = \frac{5}{84} + \frac{1}{25} = \frac{209}{2100} < \frac{1}{10},\\ \frac{1}{10} - \frac{209}{2100} &= \frac{1}{2100} > 0.\end{aligned} \tag{7.6}$$

The inequalities remain strict at $\phi = 1/100$ and $\gamma/\nu = 1/100$. No coefficient depends on phase, so this normalized estimate is uniform over the declared interval at one fixed tuple. At the excluded phase $\phi = 0$, $C = 0$; none of these normalized divisions is made there.

8 Actual receiving and net native recovery

8.1 Positive projection bounds before squaring

The reference receiver norm at t_R is $\sqrt{\beta}|z^{(0)}(t_R)| = |Z(t_R)|$. Its exact amplitude in (6.6) has the strict lower estimates

$$\begin{aligned}\frac{2\sqrt{30}}{17} > \frac{5}{8}, \qquad \frac{120}{289} - \frac{25}{64} &= \frac{455}{18496} > 0,\\ \cos(\phi/2) \ge 1 - \frac{\phi^2}{8} &\ge \frac{79999}{80000} > \frac{24}{25},\\ \frac{79999}{80000} - \frac{24}{25} = \frac{3199}{80000} > 0, \qquad \frac{\sqrt{\beta}|z^{(0)}(t_R)|}{\sqrt{2C}} &> \frac{3}{5}.\end{aligned} \tag{8.1}$$

The carrier, weighted receiver and joint external projections are contractive for the auxiliary norm. Reverse triangle and (7.6) thus give

$$\sqrt{\beta}|z(t_R)| > \left(\frac{3}{5} - \frac{1}{10}\right)\sqrt{2C} = \frac{1}{2}\sqrt{2C} > 0.$$

This lower bound is positive before squaring. The reference carrier is zero at t_R, and its entire external state is zero at t_F. With the original one-half factors in (2.4), it follows that

$$\begin{aligned} H_R(t_R) &= \tfrac{1}{2}(\sqrt{\beta}|z(t_R)|)^2 > C/4, \\ H_C(t_R) &= \tfrac{1}{2}\,\|(p, r_c)(t_R)\|_2^2 < C/100, \\ E_F &= \tfrac{1}{2}\,\|(p, r_c, z)(t_F)\|_{CR}^2 < C/100, \end{aligned} \tag{8.2}$$

where $\|(p, r_c, z)\|_{CR}^2 = \|p\|_2^2 + \|r_c\|_2^2 + \beta|z|^2$.

8.2 Inherited access on these same coefficients

For the fixed actual coefficients selected in (3.2),

$$\begin{aligned} \frac{\sigma_{\mathrm{ext}}^2}{\nu^2} = \left(\frac{\gamma}{\nu}\right)^2 + \frac{1}{9} \le \frac{1}{10000} + \frac{1}{9} < \frac{1}{4}, \\ \frac{1}{4} - \left(\frac{1}{10000} + \frac{1}{9}\right) = \frac{12491}{90000} > 0. \end{aligned} \tag{8.3}$$

All remaining P4 hypotheses hold: the same fixed yaw, admitted profiles and finite positive coefficients, normalized coherent empty launch, and $\phi \in (0, \pi/12)$. Its Theorem 3.1 therefore supplies $E_A > 1053C/4096$ on this same actual solution at $t_A = 1/\nu$. It is an inherited access statement, not an additional theorem proved by the present comparison.

8.3 Separate denominators and exact recovery accounts

Before taking any ratio, we have $C > 0$, $H_R(t_R) > C/4 > 0$, $E_R \ge H_R(t_R) > C/4 > 0$, and $E_A > 1053C/4096 > 0$. In contrast E_F need only be nonnegative. Using $H_C(t_R) \ge 0$ and $0 \le H_R(t_F) \le E_F$ gives

$$\begin{aligned} \frac{H_C(t_R)}{H_R(t_R)} < \frac{1}{25}, \qquad \frac{H_R(t_R)}{E_R} = \frac{1}{1 + H_C(t_R)/H_R(t_R)} > \frac{25}{26}, \\ 0 \le \frac{H_R(t_F)}{H_R(t_R)} < \frac{1}{25}. \end{aligned} \tag{8.4}$$

The receiving share is at most one. Conservation (2.7) gives, with these positive denominators,

$$\begin{aligned} \mathcal{R}_A = 1 - \frac{E_F}{E_A} > 1 - \frac{4096}{105300} = \frac{25301}{26325} > \frac{24}{25}, \\ \frac{25301}{26325} - \frac{24}{25} = \frac{29}{26325} > 0, \\ \mathcal{R}_R = 1 - \frac{E_F}{E_R} > 1 - \frac{1}{25} = \frac{24}{25}, \\ H_N(t_F) - H_N(t_R) = E_R - E_F > \left(\frac{1}{4} - \frac{1}{100}\right) C = \frac{6}{25} C. \end{aligned} \tag{8.5}$$

Both recovery fractions are at most one because $E_F \ge 0$. Together with the chronology in Section 7.1, this completes the proof of Theorem 3.1. The sharper intermediate arithmetic for $\mathcal{R}_A$ is retained from the accepted proof, without optimizing the stated result.

Table 1: Distinct numerators, denominators and meanings.

Account	Denominator	Meaning
$H_R(t_R)$	C	Receiver stock relative to joint capacity
$H_R(t_R)$	E_R	Receiver share of intermediate external stock
$H_R(t_F)$	$H_R(t_R)$	Receiver stock remaining at the later time
$H_N(t_F) - H_N(t_A)$	E_A	Net native recovery from the access time
$H_N(t_F) - H_N(t_R)$	E_R	Net native recovery from the receiving time

8.4 What the integrated power accounts imply

Equation (2.7) gives positive integrated receiver supply on $(0, t_R)$, negative integrated receiver supply on (t_R, t_F), and negative integrated native supply on (t_R, t_F). Indeed $H_R(t_R) > 0$, $H_R(t_F) < H_R(t_R)$ and $H_N(t_F) > H_N(t_R)$ follow from the proved stocks. Continuity of each power then guarantees a nonempty open interval of its corresponding strict sign within the specified window.

Those receiver and native intervals need not coincide. The integrated inequalities do not establish exact-time power signs at t_R or t_F, a first return, an exact stock maximum or a sustained monotone receiving/recovery window.

9 A nonempty finite positive parameter family

Fix any admitted finite geometry and positive finite c, α, β, G. Then k, m, d, R are finite and positive. Choose a finite $\nu_{\text{target}} \geq 100c\sqrt{d/m} > 0$, and set once, before phase,

$$\eta = \frac{\nu_{\text{target}}\sqrt{2\alpha}}{kR\sqrt{m}} > 0, \qquad \kappa = \frac{\nu_{\text{target}}}{3}\sqrt{\frac{\beta}{m}} > 0. \tag{9.1}$$

Substitution yields $\nu = \nu_{\text{target}}$, $\omega_R = \nu/3$ and $0 < \gamma/\nu \leq 1/100$. These coefficients remain constant on every trajectory. The construction involves no zero-speed actual system, infinite-coupling limit, profile change, receiver redesign, reset or added feedback.

The equality $\omega_R = \nu/3$ is material. This is an equality-constrained sufficient subfamily, not a proved open neighborhood of robust parameter values or a classification of all tuples allowed by P4. Appendix A restores native amplitude units and verifies that both selected couplings are dimensionless.

10 Meaning, evidence and continuation

10.1 Capacity-sized does not mean whole-native-sized

The result links three events on one unchanged solution: P4-certified external access, a later receiver stock larger than a quarter of the joint capacity, and subsequent quantified net native

recovery. Its denominators are specified in Table 1. In particular,

$$\frac{C}{H_N(0)} = \frac{2}{3}(1-\cos\phi) \longrightarrow 0 \qquad \text{as } \phi \downarrow 0. \tag{10.1}$$

The ceiling $E_{\text{out}}(t) \le C$ implies that both actual external stocks and the positive net recovered amounts in (8.5) are at most C. Absolute stocks and recovered amounts can therefore tend to zero while the fractional guarantees remain uniform. The small phase interval is an essential restriction.

There is no claim of ceiling saturation, large whole-native transfer, an exact maximizing time, first return, nonlinear recurrence or period. The accounts are net changes rather than tagged-cohort transport or sustained monotonicity. The receiver is the original collocated conservative receiver, not an irreversible load, and it has no assigned material location. Trial-storage commensurability does not constitute physical calibration.

10.2 Evidence and contribution status

The accepted investigation and its actual GPT review are retained unchanged [5, 6]. The investigation's fourteen exact-algebra groups were independently replayed during that internal review, with matching group results. A replay of those same fourteen groups is not twenty-eight separate findings. The written arguments handle regularity, periodic integration, zero-norm regularization, the scalar majorant, strictness and denominators; symbolic checks are not a machine-formal proof of these analytical steps.

The present work is manuscript preparation from accepted mathematics. GPT formulated and internally reviewed the investigation; Codex independently investigated it and prepared this manuscript under the research author's direction. The new PDF and proposed payload await GPT/author manuscript review. No external peer review is claimed, no scientific suite is rerun for typesetting, and retained source records are not described as fresh remote verification.

10.3 Geometric distinctions and the next agreed direction

The energy account does not reconstruct a spatial circuit. The programme retains the central starting/meeting region, opposed cones, outward spreading, curved around-and-inward return, stationary supports and possible transfer elsewhere as geometric directions. The adopted carrier circle and intended tight toroid remain distinct, as do the close original enclosing sphere and extended-line sphere. Finite host-notch N regions are not the three-upper/three-lower central G boundaries. The tentative 15–20%, possibly more, remains author intuition rather than a coefficient or denominator.

The bounded receiving/recovery investigation is closed at its accepted scope. Broader phase/rate coverage, optimal transfer, saturation and physical/spatial calibration remain open questions, not defects to repair in this certificate. After manuscript review and separately authorized publication closeout, the agreed next scientific assessment is native F stepping and exchange cadence, distinguishing the exact native baseline increment from exchange debit or credit. No native F jump, cadence or physical clock enters this theorem.

Intended central-route correspondence and six-opening transport follow that direction. Gravity–magnetism stays linked across the programme; outer geometry, distinct golden-ratio relationships and further justified SRG integration remain later directions. Three-Way remains parked. Pi and Hóki póki brainstorming supply no premise here; the elementary inequalities in (7.1) are only arithmetic tools. This discussion commissions no successor investigation.

A Dimensions and restored native amplitude

Use P1's abstract carrier units $\mathrm{U}, \mathrm{L}, \mathrm{T}$ and native amplitude unit A [1, 4]. The normalized numeric launch uses a unit amplitude A_* with numeric value one. Restoring it gives

$$\begin{gathered} v = A_*(1,1,1), \quad \Omega_0 = A_*(1, e^{i\phi}, 1), \quad R = \sqrt{3}\, A_*, \\ D = K[v]_\times, \qquad |\ell|^2 = \frac{20G^2 A_*^2}{7}, \\ C = \alpha A_*^2(1-\cos\phi), \qquad X_0 = -\sqrt{\alpha}\, A_* \sin\phi, \qquad y_0 = 2\sqrt{C}. \end{gathered} \tag{A.1}$$

All initial deviations acquire the same A_* factor; the directions e, n stay dimensionless. Retaining R^2 in ν^2 and the amplitude of v in D preserves the dimensionless identity $\omega_N^2/\nu^2 = 5/6$.

Table 2: Dimensions of the original and auxiliary quantities.

Quantity	Dimension
a, b, u, v, R, A_*	A
K, k, G	$\mathrm{UL}/(\mathrm{TA}^2)$
D, ℓ, J	$\mathrm{UL}/(\mathrm{TA})$
p, r_c, y	U
f, z	UL/T
$\alpha;\ \beta;\ \eta, \kappa$	$\mathrm{U}^2\mathrm{L}/\mathrm{A}^2;\ \mathrm{T}^2/\mathrm{L};\ 1$
$m;\ d;\ c$	$\mathrm{L}^{-1};\ \mathrm{L}^{-3};\ \mathrm{L}/\mathrm{T}$
$\|Y\|_*, X, P, Z, \sqrt{C}$	$\mathrm{U}\sqrt{\mathrm{L}}$
L (nonlinear-bound coefficient)	$(\mathrm{U}\sqrt{\mathrm{L}}\,\mathrm{T})^{-1}$
$\gamma, \nu, \omega_N, \omega_R, \varpi$	T^{-1}
H_N, H_C, H_R, C	$\mathrm{U}^2\mathrm{L}$
θ and all stock fractions	1

The coefficient L is distinct from the length unit L and the circle length L_c. The dimensionally restored identity used in (7.2) is

$$\theta = \frac{\sqrt{2}\, A_*}{R}\sqrt{1-\cos\phi}\,\nu T \le \frac{\phi}{\sqrt{3}}\nu T < \frac{1}{25}. \tag{A.2}$$

The factor $A_*/R = 1/\sqrt{3}$ must not be suppressed in a dimensional identity. Both choices in (9.1) are dimensionless couplings. Restoring amplitude changes none of the normalized fractions or finite-error bounds. The reference frequencies are analytical quantities, not a physical clock or native-step cadence.

References

[1] H. Frímann Halldórsson, *A Reciprocal Chiral Exchange Extension of the Tri-Octagon Model*, v0.1 (2026). Immutable published edition and evidence. Theorem 3.1 supplies global well-posedness and strong empty-launch regularity.

[2] H. Frímann Halldórsson, *Same-Trajectory Native Credit in the Reciprocal Chiral Exchange Candidate*, v0.1 (2026). Immutable published edition and evidence. Theorem 3.1 is the separate earlier native-credit theorem; Section 5 gives comparison-method precedent.

[3] H. Frímann Halldórsson, *Recoverable Transfer and Projection Capacity in the Reciprocal Chiral Exchange Candidate*, v0.1 (2026). Immutable published edition and evidence. Theorem 3.1 supplies the projection ceiling; Theorem 4.1 uses a distinct receiving/recovery threshold.

[4] H. Frímann Halldórsson, *Finite-Time Access to Native Projection Capacity in the Reciprocal Chiral Exchange Candidate*, v0.1 (2026). Immutable published edition and evidence. Theorem 3.1 supplies access at $1/\nu$ after its rate condition is checked.

[5] H. Frímann Halldórsson, with GPT formulation and Codex independent investigation, *Native capacity receiving and recovery: independent proof*, 5 November 2026. Accepted proof, claim/source ledger and fourteen-group exact-algebra evidence, preserved with this manuscript.

[6] GPT, *Review of Capacity-Sized Receiving and Native Recovery*, 5 November 2026. Project-internal mathematical and evidence review: investigation accepted unchanged, with independent replay of the same fourteen algebra groups. This acceptance does not extend to the new typeset manuscript.